Razeen Wasif

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Summary

Software Engineer & ML Researcher with a strong foundation in **AI, machine learning,** and **software construction**. My background in physics provides a rigorous analytical and quantitative approach to problem-solving. Experienced in designing and implementing AI models, building **training infrastructure**, and shipping **full-stack platforms** using **Python, C++, CUDA,** and **TypeScript**.

Professional Experience

ANU Virtuous Loop

Full-Stack Developer & UX Lead Feb 2026 – Present

- Collaborating with an industry client to architect a robust, scalable feedback dashboard using the **Microsoft Power Platform (Power Apps, Power BI, and Dataverse)**.
- Spearheading a complete **UI/UX overhaul** by conducting user research and wireframing, resulting in a 40% improvement in navigation efficiency and data accessibility.
- Engineering automated data pipelines with **Power Automate** to replace legacy manual processes, ensuring real-time feedback synchronization and system reliability.
- Implementing **Row-Level Security (RLS)** and enterprise-grade data validation logic to ensure the platform meets ANU's strict privacy and compliance standards.
- Managing the full **Software Development Life Cycle (SDLC)** in an Agile environment, including stakeholder requirement gathering, sprint planning, and client demonstrations.

Projects

Prism: GPU-Accelerated AutoML & Entity Resolution Engine

Live / Github

- Engineered a GPU-first AutoML pipeline for heterogeneous tabular data, featuring automated EDA, target-aware cleaning, and Optuna-powered hyperparameter optimisation.
- Architected a microservices stack using Docker Compose, orchestrating FastAPI backends, a React/Vite frontend, and an Ollama-based LLM server for natural language data exploration.
- Integrated a RAPIDS-accelerated record linkage workflow for entity resolution, utilising asynchronous execution and semantic embeddings to maximise candidate pair recall.
- Developed custom CUDA-backed XGBoost and PyTorch model wrappers with automated hardware fallback (CUDA/MPS/CPU), optimising training throughput for large-scale datasets.

Nexus: Distributed Training Console

Live

- Architected a standalone control plane and real-time monitoring console for distributed ML training, with a self-contained backend, brand, and deployment surface.
- Built a **Firestore-backed** pod inventory and run-lifecycle model, surfacing NCCL liveness probes and per-rank telemetry rolled up live in a monochrome glass shell.
- Designed a streaming event console for multi-rank orchestration, exposing per-step throughput, gradient norms, and rank-level failures without polling.
- Shipped on **React + TypeScript + Firebase** with a Python ingestion service, separating the operator UI from the training cluster's runtime.

Council: Multi-Model Coding Agent

Github

Education

Australian National University

Bachelor of Advanced Computing (Honors)

- **Coursework:** Machine Learning, Artificial Intelligence, Computer Vision, Software Engineering, Computer Architecture, Algorithms and Data structures, Data analysis

Open Source Contributions

Pygame Library (GitHub)

- Identified and resolved a critical **segmentation fault** within the `PixelArray` module, enhancing library stability.
- Debugged memory management issues in the **C-based backend** of the library to prevent invalid memory access during array slicing operations.

Research

Partial Fine-Tuning of BioCLIP 2.5 for Multi-Label Plant ID (PlantCLEF 2026)

Project / Kaggle

- Partial fine-tune of **BioCLIP 2.5 ViT-H/14** with auxiliary **genus / family / order / class** heads, injecting taxonomic structure into the loss to stabilise training on a 7,806-class long-tail benchmark.
- Engineered a **4×4 tile inference** pipeline ensembled across **224 and 336-pixel** resolutions, with **logit-adjusted adaptive thresholding** per class to handle multi-label vegetation plots.
- Achieved **0.418 Macro F1** on the PlantCLEF 2026 leaderboard; paper submitted to **LifeCLEF / CLEF 2026** (ANU at PlantCLEF 2026).

Signal-Preserving TinyML for Gravitational Wave Triggers

github

- Investigated the impact of **INT8 quantization** on phase-coherence in Gravitational Wave (GW) signals for deployment on ultra-low-power **ARM**

- Designed a seven-member AI council layered on a coding CLI — **Architect, Implementer, Skeptic, Critic, Tester, Security,** and **Performance** lenses propose in parallel before a synthesizer reduces to a single plan.
- Implemented a single-executor diff pipeline so only one role writes code while the others vote **pass / nit / concern / block**, with a single revision pass triggered on three or more blocks.
- Built the runtime in **TypeScript on Bun / Node.js** with a multi-model backend, fanning out role prompts in parallel and aggregating structured votes for gating.

Oracle: Multi-Domain Identification Platform Live

- Engineered a live multi-domain identification platform that productizes the **BioCLIP 2.5** PlantCLEF model — image upload, top-k predictions with taxonomy context, served via a **React + TypeScript + Firebase** front end.
- Implemented the inference path with **4×4 tile decomposition** and multi-resolution (224 / 336 px) ensembling, with logit-adjusted adaptive thresholding tuned for long-tail multi-label outputs.
- Validated end-to-end on the **PlantCLEF 2026** benchmark (7,806-class long-tail), reaching **0.418 Macro F1** on the held-out vegetation-plot test set.

Custom 16-Bit RISC CPU Design & Implementation github

- Designed and simulated a fully functional **16-bit RISC processor** from basic logic gates, featuring a custom Instruction Set Architecture (ISA) with support for arithmetic, logic, and control-flow operations.
- Engineered a **Multi-Stage Datapath** incorporating an Arithmetic Logic Unit (ALU), a Register File, and a microprogrammed **Control Unit** to handle complex instruction decoding and execution cycles.
- Optimized memory management by implementing a 16-bit address bus with dedicated **Program Counter (PC)** and **Instruction Registers (IR)**, facilitating efficient fetch-decode-execute loops.
- Validated hardware logic through rigorous unit testing of individual components (ALU, MUX, Decoders) and end-to-end simulation of assembly-level programs (e.g., Fibonacci, array traversal).

High-Performance Social Media Backend & Mobile Application github

- Developed a scalable Java-based backend for an Android social media application, implementing complex features like real-time message reactions, search, and content moderation.
- Engineered a custom **AVL Tree** data structure from scratch to ensure $O(\log n)$ performance for data retrieval, meeting strict latency requirements (1M operations/sec) for reaction processing.
- Designed a robust **Censorship Facade** with **Leet Speak decoding** and pattern matching to identify and filter profanity through custom normalization algorithms.
- Leveraged advanced **Design Patterns** including **DAO, Singleton, Factory, and Template** to build a modular, extensible, and SOLID-compliant architecture.
- Optimized a **persistence layer** using JSON and binary formats, achieving high space efficiency (<40 bytes/record) and high-speed I/O (100k records/sec).
- Refactored legacy codebases to eliminate code smells and improve maintainability, achieving high statement coverage through **JUnit parameterized and black-box testing**.

Cortex-M hardware.

- Developed a "Signal-Preserving" framework using **Quantization-Aware Training (QAT)** and Hardware-Aware Distillation to minimize SNR loss in stochastic astrophysical data.
- Engineered custom **C++ inference kernels** to bypass TensorFlow Lite Micro overhead, optimizing for high-frequency (4096 Hz) strain data within a **¡128KB SRAM** budget.
- Balanced sensitivity against False Alarm Rates to achieve over 99% data compression on edge devices for autonomous astronomical sensors.

Technologies

Languages: Python, C++, CUDA, Java, TypeScript, ARM ASM, SQL

ML / AI: PyTorch, TensorFlow, Optuna, RAPIDS, BioCLIP, DINOv3

Systems: Linux Kernel, TensorRT, NVIDIA DALI, FastAPI, Docker

Platforms: Microsoft Power Platform, Power BI, Dataverse, Firebase, Figma, React

Certifications

Deloitte Australia Technology Job Simulation
Forage *Aug 2024*

- Completed a job simulation involving data analysis for Deloitte's Technology team. Created a data dashboard using Tableau. Wrote a proposal for a client project to create a functioning dashboard. Used Excel to classify data and draw business conclusions

Study Australia Industry Experience Program
Practera *July 2025*

- Applied digital marketing and SEO best practices to develop a comprehensive strategy for a real estate startup, including competitor analysis and video automation recommendations.